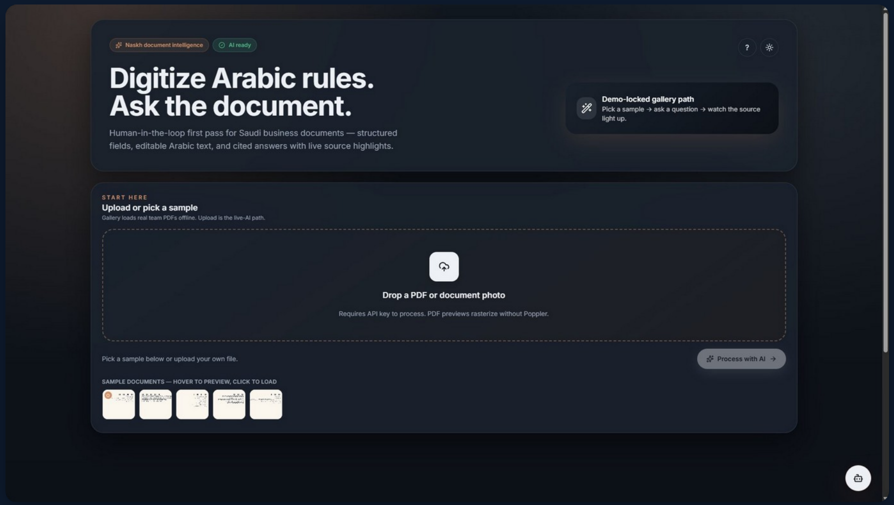
NASKH

نسخ · document intelligence

Digitize Arabic rules.
Ask the document.

A human-in-the-loop AI first pass for Saudi regulatory and business documents — structured fields, editable Arabic text, and cited answers with live source highlights.



Live source highlighting on real Saudi documents

WeCloudData 2026 Hackathon · Saudi Digital Academy

Team Naskh

THE PROBLEM

Saudi business runs on documents AI can't read.

Locked in paper and PDFs

Regulatory circulars, contracts, MoUs and board minutes live as scans, photos, and handwritten notes — not data.

Arabic breaks generic tools

RTL text, ligatures, handwriting, and bilingual Arabic/English layouts defeat mainstream OCR and off-the-shelf AI.

Manual, slow, no audit trail

Teams retype and re-read by hand. Errors slip through, and there's no link back to the source for review.

WHO FEELS IT

- Compliance and legal teams tracking obligations and deadlines
- Government-facing organizations digitizing regulatory archives
- Back-offices converting paper records into searchable data

MOTIVATION

Vision 2030 is digital — the paperwork isn't, and it's mostly Arabic.

01

A real, local backlog

Saudi Arabia's digitization push leaves mountains of paper and scanned records still waiting to become usable data.

02

Arabic is underserved

Most document-AI tooling is English-first. Arabic regulatory documents are exactly where generic solutions fall short.

03

The stakes are high

A missed clause, deadline, or obligation carries legal and financial risk — accuracy and traceability matter.

Arabic-first

Built for RTL, bilingual, and handwritten Saudi documents.

Cited & auditable

Every field and answer links back to its exact source.

Honest first pass

AI drafts, a human reviews and approves before archive.

THE SOLUTION

Naskh: a guided first pass from scan to structured, cited data.

Upload

Drop a PDF or document photo — or pick a prepared sample.

Extract

AI vision returns structured fields, confidence, and full Arabic transcription.

Review

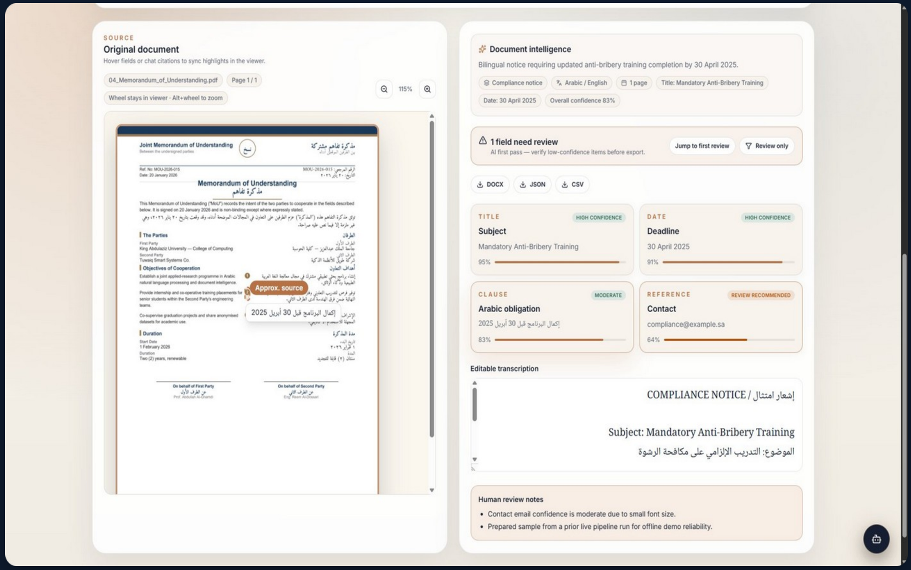
Hover a field and the exact source region lights up for verification.

Ask

A cited assistant answers questions grounded in the document via RAG.

Export

Download clean DOCX, JSON, or CSV for archive and downstream systems.



Structured fields, confidence, and review flags — with the source on the left.

ARCHITECTURE & AI

A real pipeline — server-side AI, grounded answers.

React + Vite
Tailwind UI

→

FastAPI
REST + SSE

→

OpenAI Vision
structured outputs

→

ChromaDB
per-doc RAG

API keys stay server-side — never in the browser.

Vision extraction

Pydantic-typed structured outputs: fields, confidence, and Arabic transcription in one schema.

Grounded RAG chat

Per-document embeddings in ChromaDB power streaming, cited answers — not generic ChatGPT.

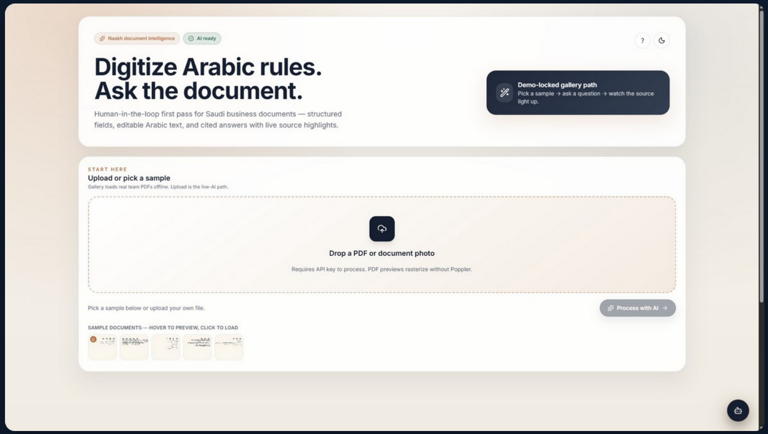
Source highlighting

Answers map to exact text-layer coordinates, lighting up the real region on the page.

React 19 · FastAPI · OpenAI gpt-4o · Pydantic · ChromaDB · pypdfium2 · arabic-reshaper + bidi · python-docx

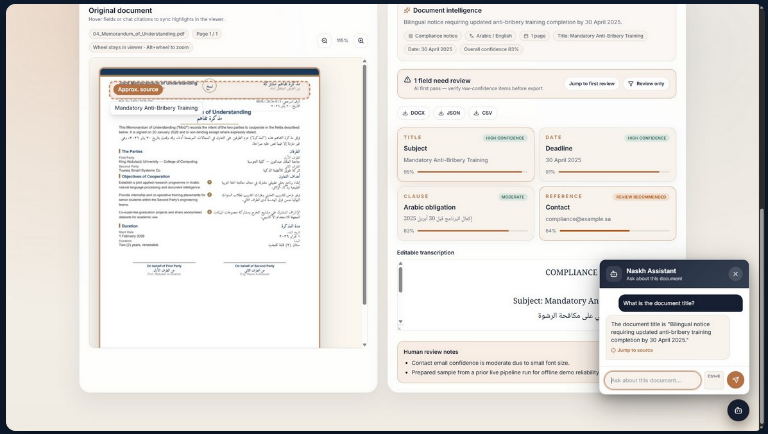
LIVE DEMO

Let's run it on a real Saudi document.



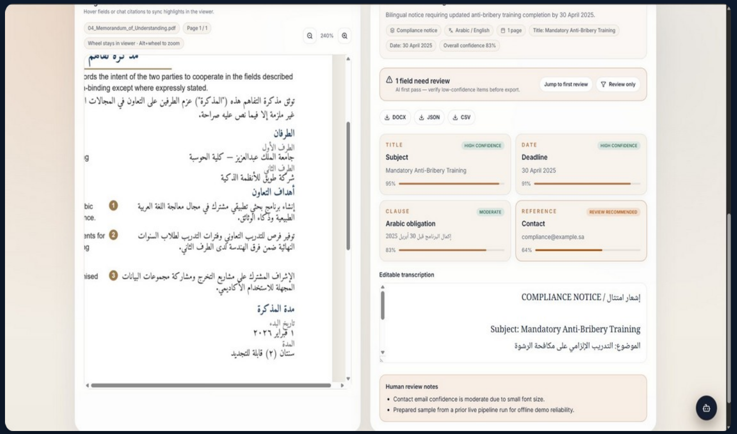
Pick & process

Light or dark, a sample loads instantly — offline-safe.



Cited assistant

Ask a question; the answer cites and jumps to source.



Inspect closely

Zoom and pan the page to verify every detail.

What you'll see: pick a sample → structured extraction with confidence → hover a field and watch the source light up → ask a cited question → export DOCX / JSON / CSV.

ROADMAP

From hackathon MVP to a real product.

NOW
Improve

- Fine-tune an Arabic handwriting model
- Raise extraction accuracy on dense layouts
- Persist operator edits across sessions

NEXT
Build next

- Batch / queue digitization at volume
- Multi-document compare and search
- Saved review workflows and roles

SCALE
Productize

- Integrations: gov portals & ECM
- Auth, audit logs, data residency
- Workers + database + object storage

Build something useful.
Build something intelligent.

Naskh solves a real Saudi problem with meaningful AI, a working MVP, a clear value story, and a realistic path to production.

Real problem | Meaningful AI | Working MVP | Cited & auditable | Production vision